Formal component-wise ablation: five backbones, six configurations (tau = 0.4 / 0.5 / 0.6)

Val split: 465 queries (255 positive / 210 negative); all values: %. Parentheses: gain versus the same-backbone GMR, pp.

		Operating threshold tau = 0.4									**Operating threshold tau = 0.5**									**Operating threshold tau = 0.6**								
		Null-Set Rejection			*Temporal Localization*				*Overall GMR*		*Null-Set Rejection*			*Temporal Localization*				*Overall GMR*		*Null-Set Rejection*			*Temporal Localization*				*Overall GMR*	
Backbone	**Configuration (cumulative)**	N-acc	AUROC	Rej-F1	mAP	mR@1	mR@5	mR+@5	G-mIoU@1	G-mIoU@3	N-acc	AUROC	Rej-F1	mAP	mR@1	mR@5	mR+@5	G-mIoU@1	G-mIoU@3	N-acc	AUROC	Rej-F1	mAP	mR@1	mR@5	mR+@5	G-mIoU@1	G-mIoU@3
Moment-DETR	1. GMR (coupled baseline)	0.00	69.21	0.00	7.35	4.65	10.86	**2.17**	6.08	2.60	88.57	69.21	69.40	7.35	4.65	10.86	**2.17**	44.00	41.56	**100.00**	69.21	**70.71**	7.35	4.65	10.86	**2.17**	48.68	46.33
	2. + Global Event Query	**100.00** (+100.00)	70.59 (+1.38)	**70.71** (+70.71)	7.35 (+0.00)	4.65 (+0.00)	10.86 (+0.00)	**2.17** (+0.00)	48.68 (+42.60)	46.33 (+43.73)	**100.00** (+11.43)	70.59 (+1.38)	**70.71** (+1.31)	7.35 (+0.00)	4.65 (+0.00)	10.86 (+0.00)	**2.17** (+0.00)	48.68 (+4.68)	46.33 (+4.77)	**100.00** (+0.00)	70.59 (+1.38)	**70.71** (+0.00)	7.35 (+0.00)	4.65 (+0.00)	10.86 (+0.00)	**2.17** (+0.00)	48.68 (+0.00)	46.33 (+0.00)
	3. + LoRA	**100.00** (+100.00)	70.75 (+1.54)	**70.71** (+70.71)	**7.94** (+0.59)	**5.20** (+0.55)	**11.56** (+0.70)	**2.17** (+0.00)	**48.91** (+42.83)	**46.41** (+43.81)	**100.00** (+11.43)	70.75 (+1.54)	**70.71** (+1.31)	**7.94** (+0.59)	**5.20** (+0.55)	**11.56** (+0.70)	**2.17** (+0.00)	**48.91** (+4.91)	**46.41** (+4.85)	**100.00** (+0.00)	70.75 (+1.54)	**70.71** (+0.00)	**7.94** (+0.59)	**5.20** (+0.55)	**11.56** (+0.70)	**2.17** (+0.00)	**48.91** (+0.23)	**46.41** (+0.08)
	4. + Hard-RoI (no saliency)	**100.00** (+100.00)	70.71 (+1.50)	**70.71** (+70.71)	6.41 (-0.94)	2.95 (-1.70)	10.69 (-0.17)	0.44 (-1.73)	48.76 (+42.68)	46.36 (+43.76)	**100.00** (+11.43)	70.71 (+1.50)	**70.71** (+1.31)	6.41 (-0.94)	2.95 (-1.70)	10.69 (-0.17)	0.44 (-1.73)	48.76 (+4.76)	46.36 (+4.80)	**100.00** (+0.00)	70.71 (+1.50)	**70.71** (+0.00)	6.41 (-0.94)	2.95 (-1.70)	10.69 (-0.17)	0.44 (-1.73)	48.76 (+0.08)	46.36 (+0.03)
	5. + Gaussian-RoI (no saliency)	**100.00** (+100.00)	71.05 (+1.84)	**70.71** (+70.71)	7.41 (+0.06)	3.52 (-1.13)	11.17 (+0.31)	0.28 (-1.89)	48.61 (+42.53)	46.31 (+43.71)	**100.00** (+11.43)	71.05 (+1.84)	**70.71** (+1.31)	7.41 (+0.06)	3.52 (-1.13)	11.17 (+0.31)	0.28 (-1.89)	48.61 (+4.61)	46.31 (+4.75)	**100.00** (+0.00)	71.05 (+1.84)	**70.71** (+0.00)	7.41 (+0.06)	3.52 (-1.13)	11.17 (+0.31)	0.28 (-1.89)	48.61 (-0.07)	46.31 (-0.02)
	6. Full DeRoI-GMR (+6D saliency)	**100.00** (+100.00)	**71.07** (+1.86)	**70.71** (+70.71)	7.01 (-0.34)	3.60 (-1.05)	10.23 (-0.63)	0.28 (-1.89)	48.70 (+42.62)	46.34 (+43.74)	**100.00** (+11.43)	**71.07** (+1.86)	**70.71** (+1.31)	7.01 (-0.34)	3.60 (-1.05)	10.23 (-0.63)	0.28 (-1.89)	48.70 (+4.70)	46.34 (+4.78)	**100.00** (+0.00)	**71.07** (+1.86)	**70.71** (+0.00)	7.01 (-0.34)	3.60 (-1.05)	10.23 (-0.63)	0.28 (-1.89)	48.70 (+0.02)	46.34 (+0.01)
QD-DETR	1. GMR (coupled baseline)	56.19	**74.15**	59.60	**19.60**	**13.30**	**26.09**	10.22	36.21	30.79	80.95	**74.15**	69.11	**19.60**	**13.30**	**26.09**	10.22	45.16	40.78	92.38	**74.15**	**70.67**	**19.60**	**13.30**	**26.09**	10.22	49.52	44.73
	2. + Global Event Query	81.90 (+25.71)	73.81 (-0.34)	68.39 (+8.79)	**19.60** (+0.00)	**13.30** (+0.00)	**26.09** (+0.00)	10.22 (+0.00)	45.99 (+9.78)	40.98 (+10.19)	92.38 (+11.43)	73.81 (-0.34)	70.16 (+1.05)	**19.60** (+0.00)	**13.30** (+0.00)	**26.09** (+0.00)	10.22 (+0.00)	49.44 (+4.28)	44.73 (+3.95)	**96.19** (+3.81)	73.81 (-0.34)	70.63 (-0.04)	**19.60** (+0.00)	**13.30** (+0.00)	**26.09** (+0.00)	10.22 (+0.00)	**50.74** (+1.22)	**46.14** (+1.41)
	3. + LoRA	81.43 (+25.24)	73.73 (-0.42)	68.40 (+8.80)	17.79 (-1.81)	11.48 (-1.82)	23.77 (-2.32)	**11.48** (+1.26)	45.11 (+8.90)	40.67 (+9.88)	92.38 (+11.43)	73.73 (-0.42)	70.29 (+1.18)	17.79 (-1.81)	11.48 (-1.82)	23.77 (-2.32)	**11.48** (+1.26)	48.83 (+3.67)	44.50 (+3.72)	95.24 (+2.86)	73.73 (-0.42)	70.18 (-0.49)	17.79 (-1.81)	11.48 (-1.82)	23.77 (-2.32)	**11.48** (+1.26)	49.79 (+0.27)	45.50 (+0.77)
	4. + Hard-RoI (no saliency)	**85.71** (+29.52)	73.44 (-0.71)	69.36 (+9.76)	18.31 (-1.29)	11.90 (-1.40)	25.28 (-0.81)	9.78 (-0.44)	**47.37** (+11.16)	**42.52** (+11.73)	**93.81** (+12.86)	73.44 (-0.71)	70.23 (+1.12)	18.31 (-1.29)	11.90 (-1.40)	25.28 (-0.81)	9.78 (-0.44)	50.05 (+4.89)	**45.34** (+4.56)	**96.19** (+3.81)	73.44 (-0.71)	70.26 (-0.41)	18.31 (-1.29)	11.90 (-1.40)	25.28 (-0.81)	9.78 (-0.44)	50.53 (+1.01)	46.05 (+1.32)
	5. + Gaussian-RoI (no saliency)	**85.71** (+29.52)	73.51 (-0.64)	**69.50** (+9.90)	17.89 (-1.71)	12.11 (-1.19)	24.72 (-1.37)	6.93 (-3.29)	47.01 (+10.80)	42.32 (+11.53)	**93.81** (+12.86)	73.51 (-0.64)	70.23 (+1.12)	17.89 (-1.71)	12.11 (-1.19)	24.72 (-1.37)	6.93 (-3.29)	**50.06** (+4.90)	45.21 (+4.43)	**96.19** (+3.81)	73.51 (-0.64)	70.26 (-0.41)	17.89 (-1.71)	12.11 (-1.19)	24.72 (-1.37)	6.93 (-3.29)	50.58 (+1.06)	46.01 (+1.28)
	6. Full DeRoI-GMR (+6D saliency)	83.81 (+27.62)	73.82 (-0.33)	69.02 (+9.42)	18.02 (-1.58)	11.05 (-2.25)	25.84 (-0.25)	7.54 (-2.68)	45.83 (+9.62)	41.34 (+10.55)	93.33 (+12.38)	73.82 (-0.33)	**70.63** (+1.52)	18.02 (-1.58)	11.05 (-2.25)	25.84 (-0.25)	7.54 (-2.68)	49.64 (+4.48)	45.06 (+4.28)	95.71 (+3.33)	73.82 (-0.33)	70.28 (-0.39)	18.02 (-1.58)	11.05 (-2.25)	25.84 (-0.25)	7.54 (-2.68)	50.20 (+0.68)	45.82 (+1.09)
CG-DETR	1. GMR (coupled baseline)	19.52	**74.74**	30.71	20.98	14.08	28.47	11.76	22.34	15.85	50.48	**74.74**	56.84	20.98	14.08	28.47	11.76	34.44	28.71	80.48	**74.74**	69.12	20.98	14.08	28.47	11.76	46.35	40.72
	2. + Global Event Query	89.52 (+70.00)	73.79 (-0.95)	70.54 (+39.83)	20.98 (+0.00)	14.08 (+0.00)	28.47 (+0.00)	11.76 (+0.00)	48.46 (+26.12)	43.92 (+28.07)	97.14 (+46.66)	73.79 (-0.95)	71.33 (+14.49)	20.98 (+0.00)	14.08 (+0.00)	28.47 (+0.00)	11.76 (+0.00)	51.40 (+16.96)	46.64 (+17.93)	**99.52** (+19.04)	73.79 (-0.95)	71.45 (+2.33)	20.98 (+0.00)	14.08 (+0.00)	28.47 (+0.00)	11.76 (+0.00)	51.90 (+5.55)	47.40 (+6.68)
	3. + LoRA	89.52	73.87	70.54	20.65	13.65	28.20	12.31	48.39	43.87	**97.62**	73.87	**71.93**	20.65	13.65	28.20	12.31	51.51	**46.86**	**99.52**	73.87	71.45	20.65	13.65	28.20	12.31	51.80	47.36
	4. + Hard-RoI (no saliency)	89.52 (+70.00)	73.90 (-0.84)	70.54 (+39.83)	**21.91** (+0.93)	**14.82** (+0.74)	**29.65** (+1.18)	**14.15** (+2.39)	48.68 (+26.34)	43.98 (+28.13)	**97.62** (+47.14)	73.90 (-0.84)	71.80 (+14.96)	**21.91** (+0.93)	**14.82** (+0.74)	**29.65** (+1.18)	**14.15** (+2.39)	**51.59** (+17.15)	46.84 (+18.13)	**99.52** (+19.04)	73.90 (-0.84)	71.45 (+2.33)	**21.91** (+0.93)	**14.82** (+0.74)	**29.65** (+1.18)	**14.15** (+2.39)	**52.07** (+5.72)	**47.50** (+6.78)
	5. + Gaussian-RoI (no saliency)	84.29 (+64.77)	73.83 (-0.91)	69.01 (+38.30)	18.61 (-2.37)	11.38 (-2.70)	26.35 (-2.12)	10.39 (-1.37)	45.87 (+23.53)	41.60 (+25.75)	96.19 (+45.71)	73.83 (-0.91)	71.50 (+14.66)	18.61 (-2.37)	11.38 (-2.70)	26.35 (-2.12)	10.39 (-1.37)	50.33 (+15.89)	46.07 (+17.36)	**99.52** (+19.04)	73.83 (-0.91)	**71.58** (+2.46)	18.61 (-2.37)	11.38 (-2.70)	26.35 (-2.12)	10.39 (-1.37)	51.69 (+5.34)	47.36 (+6.64)
	6. Full DeRoI-GMR (+6D saliency)	**90.95** (+71.43)	73.89 (-0.85)	**71.27** (+40.56)	21.16 (+0.18)	14.47 (+0.39)	28.61 (+0.14)	12.98 (+1.22)	**49.28** (+26.94)	**44.45** (+28.60)	97.14 (+46.66)	73.89 (-0.85)	71.08 (+14.24)	21.16 (+0.18)	14.47 (+0.39)	28.61 (+0.14)	12.98 (+1.22)	51.07 (+16.63)	46.45 (+17.74)	**99.52** (+19.04)	73.89 (-0.85)	71.45 (+2.33)	21.16 (+0.18)	14.47 (+0.39)	28.61 (+0.14)	12.98 (+1.22)	51.81 (+5.46)	47.41 (+6.69)
EaTR	1. GMR (coupled baseline)	50.95	**75.63**	57.84	20.26	13.78	26.75	12.44	34.76	28.83	69.05	**75.63**	65.02	20.26	13.78	26.75	12.44	41.62	36.00	87.62	**75.63**	69.96	20.26	13.78	26.75	12.44	48.47	43.35
	2. + Global Event Query	85.24 (+34.29)	72.46 (-3.17)	68.45 (+10.61)	20.26 (+0.00)	13.78 (+0.00)	26.75 (+0.00)	12.44 (+0.00)	47.11 (+12.35)	42.38 (+13.55)	96.19 (+27.14)	72.46 (-3.17)	71.00 (+5.98)	20.26 (+0.00)	13.78 (+0.00)	26.75 (+0.00)	12.44 (+0.00)	51.17 (+9.55)	46.23 (+10.23)	**99.05** (+11.43)	72.46 (-3.17)	71.11 (+1.15)	20.26 (+0.00)	13.78 (+0.00)	26.75 (+0.00)	12.44 (+0.00)	51.71 (+3.24)	47.23 (+3.88)
	3. + LoRA	86.67 (+35.72)	72.76 (-2.87)	**68.94** (+11.10)	**20.66** (+0.40)	**14.27** (+0.49)	27.25 (+0.50)	**12.89** (+0.45)	**48.07** (+13.31)	**43.14** (+14.31)	96.19 (+27.14)	72.76 (-2.87)	70.88 (+5.86)	**20.66** (+0.40)	**14.27** (+0.49)	27.25 (+0.50)	**12.89** (+0.45)	51.17 (+9.55)	46.23 (+10.23)	**99.05** (+11.43)	72.76 (-2.87)	**71.23** (+1.27)	**20.66** (+0.40)	**14.27** (+0.49)	27.25 (+0.50)	**12.89** (+0.45)	51.75 (+3.28)	47.24 (+3.89)
	4. + Hard-RoI (no saliency)	**87.14** (+36.19)	72.89 (-2.74)	68.93 (+11.09)	19.32 (-0.94)	13.79 (+0.01)	25.65 (-1.10)	10.44 (-2.00)	48.05 (+13.29)	42.94 (+14.11)	**97.14** (+28.09)	72.89 (-2.74)	**71.20** (+6.18)	19.32 (-0.94)	13.79 (+0.01)	25.65 (-1.10)	10.44 (-2.00)	**51.53** (+9.91)	**46.60** (+10.60)	**99.05** (+11.43)	72.89 (-2.74)	71.11 (+1.15)	19.32 (-0.94)	13.79 (+0.01)	25.65 (-1.10)	10.44 (-2.00)	51.91 (+3.44)	47.28 (+3.93)
	5. + Gaussian-RoI (no saliency)	86.19 (+35.24)	72.77 (-2.86)	68.56 (+10.72)	19.41 (-0.85)	13.00 (-0.78)	**27.42** (+0.67)	10.89 (-1.55)	47.57 (+12.81)	42.47 (+13.64)	96.67 (+27.62)	72.77 (-2.86)	70.98 (+5.96)	19.41 (-0.85)	13.00 (-0.78)	**27.42** (+0.67)	10.89 (-1.55)	51.33 (+9.71)	46.36 (+10.36)	**99.05** (+11.43)	72.77 (-2.86)	71.11 (+1.15)	19.41 (-0.85)	13.00 (-0.78)	**27.42** (+0.67)	10.89 (-1.55)	51.82 (+3.35)	47.24 (+3.89)
	6. Full DeRoI-GMR (+6D saliency)	85.71 (+34.76)	72.74 (-2.89)	68.44 (+10.60)	19.54 (-0.72)	13.39 (-0.39)	26.54 (-0.21)	11.44 (-1.00)	47.22 (+12.46)	42.23 (+13.40)	96.67 (+27.62)	72.74 (-2.89)	71.10 (+6.08)	19.54 (-0.72)	13.39 (-0.39)	26.54 (-0.21)	11.44 (-1.00)	51.32 (+9.70)	46.39 (+10.39)	**99.05** (+11.43)	72.74 (-2.89)	71.11 (+1.15)	19.54 (-0.72)	13.39 (-0.39)	26.54 (-0.21)	11.44 (-1.00)	**51.94** (+3.47)	**47.29** (+3.94)
FlashVTG	1. GMR (coupled baseline)	36.67	**74.89**	48.12	**29.16**	**18.77**	38.28	14.74	31.02	23.61	72.38	**74.89**	66.38	**29.16**	**18.77**	38.28	14.74	44.61	38.32	87.62	**74.89**	69.96	**29.16**	**18.77**	38.28	14.74	49.66	44.15
	2. + Global Event Query	88.10 (+51.43)	72.23 (-2.66)	**69.68** (+21.56)	28.96 (-0.20)	18.46 (-0.31)	38.28 (+0.00)	14.74 (+0.00)	49.56 (+18.54)	43.86 (+20.25)	94.76 (+22.38)	72.23 (-2.66)	70.69 (+4.31)	28.96 (-0.20)	18.46 (-0.31)	38.28 (+0.00)	14.74 (+0.00)	51.50 (+6.89)	46.25 (+7.93)	97.14 (+9.52)	72.23 (-2.66)	70.96 (+1.00)	28.96 (-0.20)	18.46 (-0.31)	38.28 (+0.00)	14.74 (+0.00)	52.18 (+2.52)	46.90 (+2.75)
	3. + LoRA	88.10 (+51.43)	72.66 (-2.23)	69.42 (+21.30)	28.59 (-0.57)	18.21 (-0.56)	38.44 (+0.16)	14.59 (-0.15)	49.36 (+18.34)	43.67 (+20.06)	94.29 (+21.91)	72.66 (-2.23)	70.46 (+4.08)	28.59 (-0.57)	18.21 (-0.56)	38.44 (+0.16)	14.59 (-0.15)	51.25 (+6.64)	45.96 (+7.64)	97.14 (+9.52)	72.66 (-2.23)	71.20 (+1.24)	28.59 (-0.57)	18.21 (-0.56)	38.44 (+0.16)	14.59 (-0.15)	52.31 (+2.65)	47.06 (+2.91)
	4. + Hard-RoI (no saliency)	**89.05** (+52.38)	72.18 (-2.71)	69.39 (+21.27)	27.79 (-1.37)	16.83 (-1.94)	38.37 (+0.09)	13.70 (-1.04)	49.49 (+18.47)	43.95 (+20.34)	**95.24** (+22.86)	72.18 (-2.71)	**70.92** (+4.54)	27.79 (-1.37)	16.83 (-1.94)	38.37 (+0.09)	13.70 (-1.04)	51.50 (+6.89)	46.33 (+8.01)	98.57 (+10.95)	72.18 (-2.71)	71.26 (+1.30)	27.79 (-1.37)	16.83 (-1.94)	38.37 (+0.09)	13.70 (-1.04)	52.27 (+2.61)	47.36 (+3.21)
	5. + Gaussian-RoI (no saliency)	**89.05** (+52.38)	72.15 (-2.74)	69.39 (+21.27)	27.98 (-1.18)	16.40 (-2.37)	**39.35** (+1.07)	**15.96** (+1.22)	49.59 (+18.57)	43.98 (+20.37)	**95.24** (+22.86)	72.15 (-2.74)	**70.92** (+4.54)	27.98 (-1.18)	16.40 (-2.37)	**39.35** (+1.07)	**15.96** (+1.22)	51.50 (+6.89)	46.33 (+8.01)	**99.05** (+11.43)	72.15 (-2.74)	**71.48** (+1.52)	27.98 (-1.18)	16.40 (-2.37)	**39.35** (+1.07)	**15.96** (+1.22)	**52.49** (+2.83)	**47.58** (+3.43)
	6. Full DeRoI-GMR (+6D saliency)	**89.05** (+52.38)	72.30 (-2.59)	69.39 (+21.27)	28.07 (-1.09)	16.67 (-2.10)	38.62 (+0.34)	15.43 (+0.69)	**49.69** (+18.67)	**44.02** (+20.41)	**95.24** (+22.86)	72.30 (-2.59)	**70.92** (+4.54)	28.07 (-1.09)	16.67 (-2.10)	38.62 (+0.34)	15.43 (+0.69)	**51.66** (+7.05)	**46.39** (+8.07)	98.57 (+10.95)	72.30 (-2.59)	71.26 (+1.30)	28.07 (-1.09)	16.67 (-2.10)	38.62 (+0.34)	15.43 (+0.69)	52.44 (+2.78)	47.42 (+3.27)

Formal rows only: configurations 1-6. The five fusion-enabled diagnostic runs (execution row 7) are published separately.

Added rows: value (delta versus configuration-1 GMR, same backbone / split / threshold). Gains subtract displayed two-decimal values.

Bold: within-backbone best; underline: second distinct best (ties retained). Fixed epoch 4; baseline frozen; seed 2026; no test selection.

Native FlashVTG soft score calibration / rounding / NMS is retained. Matched CG/EaTR GMR exports differ from historical mAP by 0.01/0.02 pp.